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Chapter 1

Spin vs Kramers restricted MP2

1.1 Spin-unrestricted non-relativistic MP2 energy

The non-relativistic MP2 energy in spin-orbitals, assuming Einstein summation, is

$$E_{\text{MP2}} = \frac{1}{4} \bar{t}_{ab}^{ij} \bar{g}_{ij}^{ab}. \quad (1.1)$$

Here, $\bar{g}_{ij}^{ab}$ denotes the antisymmetrized electron repulsion integrals,

$$\bar{g}_{ij}^{ab} = g_{ij}^{ab} - g_{ji}^{ab} = \langle ij|ab \rangle - \langle ji|ab \rangle = \langle ij||ab \rangle, \quad (1.2)$$

$$g_{ij}^{ab} = \langle ij|ab \rangle = \iint \phi_i^*(\mathbf{r1}) \phi_j^*(\mathbf{r2}) \frac{1}{|\mathbf{r1} - \mathbf{r2}|} \phi_a(\mathbf{r1}) \phi_b(\mathbf{r2}) d\mathbf{r1} d\mathbf{r2} \quad (1.3)$$

and

$$\bar{t}_{ab}^{ij} = \frac{\bar{g}_{ab}^{ij}}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b} = \frac{\bar{g}_{ab}^{ij}}{\Delta_{ab}^{ij}} \quad (1.4)$$

are the antisymmetrized MP2 amplitudes, where ϵ_p are the orbital energies. Indices i and j denote occupied orbitals, while indices a and b denote virtual orbitals. Note:

$$g_{ij}^{ab} = g_{ji}^{ba} = g_{ab}^{*ji} = g_{ba}^{*ji} \quad (1.5)$$

Before we look at spin-separated E_{MP2} , we need to realize that not all spin-combinations are non-zero. This can be understood by working out the spin-block separation of $\bar{g}_{ij}^{ab}$:

$$\begin{aligned} \bar{g}_{ij}^{ab} &= \bar{g}_{i\alpha j\alpha}^{a\alpha b\alpha} + \bar{g}_{i\beta j\beta}^{a\beta b\beta} + \bar{g}_{i\alpha j\beta}^{a\alpha b\beta} + \bar{g}_{i\beta j\alpha}^{a\beta b\alpha} + \bar{g}_{i\alpha j\beta}^{a\alpha b\beta} + \bar{g}_{i\alpha j\beta}^{a\beta b\alpha} \\ \bar{g}_{i\alpha j\alpha}^{a\alpha b\alpha} &= g_{i\alpha j\alpha}^{a\alpha b\alpha} - g_{j\alpha i\alpha}^{a\alpha b\alpha} \end{aligned} \quad (1.6)$$

$$\text{and } \bar{g}_{i\alpha j\beta}^{a\alpha b\beta} = g_{i\alpha j\beta}^{a\alpha b\beta} - 0 \quad (1.7)$$

$$(g_{j\beta i\alpha}^{a\alpha b\beta} \text{ is zero due to violation of column spin symmetry}) \quad (1.8)$$

Decomposing the energy expression into its individual spin components,

$$E_{\text{MP2}} = E_{\alpha\alpha}^{\alpha\alpha} + E_{\beta\beta}^{\beta\beta} + E_{\alpha\beta}^{\alpha\beta} + E_{\beta\alpha}^{\beta\alpha} + E_{\beta\alpha}^{\alpha\beta} + E_{\alpha\beta}^{\beta\alpha} \quad (1.9)$$

$$= \frac{1}{4} \left[\bar{t}_{a\alpha b\alpha}^{i\alpha j\alpha} \bar{g}_{i\alpha j\alpha}^{a\alpha b\alpha} + \bar{t}_{a\beta b\beta}^{i\beta j\beta} \bar{g}_{i\beta j\beta}^{a\beta b\beta} + \bar{t}_{a\alpha b\beta}^{i\alpha j\beta} \bar{g}_{i\alpha j\beta}^{a\alpha b\beta} + \bar{t}_{a\beta b\alpha}^{i\beta j\alpha} \bar{g}_{i\beta j\alpha}^{a\beta b\alpha} + \bar{t}_{a\beta b\alpha}^{i\alpha j\beta} \bar{g}_{i\alpha j\beta}^{a\beta b\alpha} + \bar{t}_{a\alpha b\beta}^{i\beta j\alpha} \bar{g}_{i\beta j\alpha}^{a\alpha b\beta} \right] \quad (1.10)$$

Note that $E_{\alpha\beta}^{\alpha\beta} = E_{\beta\alpha}^{\beta\alpha}$ in Eq. (1.35). This can be seen by considering the fourth term in Eq. (1.35), which, upon swapping the columns of both $\bar{t}$ and $\bar{g}$ and then swapping the dummy indices, becomes identical to the third term in Eq. (1.35). Similarly, it can also be seen that $E_{\beta\alpha}^{\alpha\beta} = E_{\alpha\beta}^{\beta\alpha}$. Consequently, Eq. (1.35) can be rewritten as

$$E_{\text{MP2}} = \frac{1}{4}\bar{t}_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} \bar{g}_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} + \frac{1}{4}\bar{t}_{a_\beta b_\beta}^{i_\beta j_\beta} \bar{g}_{i_\beta j_\beta}^{a_\beta b_\beta} + \frac{1}{2}\bar{t}_{a_\alpha b_\beta}^{i_\alpha j_\beta} \bar{g}_{i_\alpha j_\beta}^{a_\alpha b_\beta} + \frac{1}{2}\bar{t}_{a_\beta b_\alpha}^{i_\beta j_\alpha} \bar{g}_{i_\beta j_\alpha}^{a_\beta b_\alpha}. \quad (1.11)$$

Now consider the tensor network for $E_{\alpha\alpha}^{\alpha\alpha}$ only:

$$E_{\alpha\alpha}^{\alpha\alpha} = \frac{1}{4}\bar{t}_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} \bar{g}_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} \quad (1.12)$$

$$= \frac{1}{4}(t_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} - t_{b_\alpha a_\alpha}^{i_\alpha j_\alpha})(g_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} - g_{j_\alpha i_\alpha}^{a_\alpha b_\alpha}) \quad (1.13)$$

$$= \frac{1}{4}[t_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} g_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} - t_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} g_{j_\alpha i_\alpha}^{a_\alpha b_\alpha} - t_{b_\alpha a_\alpha}^{i_\alpha j_\alpha} g_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} + t_{b_\alpha a_\alpha}^{i_\alpha j_\alpha} g_{j_\alpha i_\alpha}^{a_\alpha b_\alpha}]. \quad (1.14)$$

The last term in Eq. (1.14) is equal to the first term, which can be seen by swapping the column indices of g and then exchanging the dummy indices a_α and b_α . An analogous relation holds between the second and third terms in Eq. (1.14). Therefore, the expression for $E_{\alpha\alpha}$ can be written in a compact form as

$$E_{\alpha\alpha}^{\alpha\alpha} = \frac{1}{2}[t_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} g_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} - t_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} g_{j_\alpha i_\alpha}^{a_\alpha b_\alpha}] \quad (1.15)$$

$$= \frac{1}{2}t_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} (g_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} - g_{j_\alpha i_\alpha}^{a_\alpha b_\alpha}). \quad (1.16)$$

Similarly, the expression for $E_{\beta\beta}^{\beta\beta}$ can be written in compact form as

$$E_{\beta\beta}^{\beta\beta} = \frac{1}{2}t_{a_\beta b_\beta}^{i_\beta j_\beta} (g_{i_\beta j_\beta}^{a_\beta b_\beta} - g_{j_\beta i_\beta}^{a_\beta b_\beta}). \quad (1.17)$$

Now consider $2E_{\alpha\beta}^{\alpha\beta}$:

$$2E_{\alpha\beta}^{\alpha\beta} = \frac{1}{2}\bar{t}_{a_\alpha b_\beta}^{i_\alpha j_\beta} \bar{g}_{i_\alpha j_\beta}^{a_\alpha b_\beta} \quad (1.18)$$

$$= \frac{1}{2}(t_{a_\alpha b_\beta}^{i_\alpha j_\beta} - t_{b_\beta a_\alpha}^{i_\alpha j_\beta})(g_{i_\alpha j_\beta}^{a_\alpha b_\beta} - g_{j_\beta i_\alpha}^{a_\alpha b_\beta}). \quad (1.19)$$

The second term in each bracket vanishes because it violates column spin equivalence, and hence $2E_{\alpha\beta}^{\alpha\beta}$ reduces to

$$2E_{\alpha\beta}^{\alpha\beta} = \frac{1}{2}t_{a_\alpha b_\beta}^{i_\alpha j_\beta} g_{i_\alpha j_\beta}^{a_\alpha b_\beta}. \quad (1.20)$$

Similarly, for $2E_{\beta\alpha}^{\alpha\beta}$, the first terms in the antisymmetrization brackets vanish, and we are left with

$$2E_{\beta\alpha}^{\alpha\beta} = \frac{1}{2}t_{a_\alpha b_\beta}^{i_\alpha j_\beta} g_{i_\alpha j_\beta}^{a_\alpha b_\beta}. \quad (1.21)$$

Therefore, substituting the results of Eqs. (1.16), (1.17), (1.20) and (1.21) into Eq. (1.11), we get the expression for the total unrestricted MP2 energy as:

$$E_{\text{UMP2}} = \frac{1}{2}t_{a_\alpha b_\alpha}^{i_\alpha j_\alpha} (g_{i_\alpha j_\alpha}^{a_\alpha b_\alpha} - g_{j_\alpha i_\alpha}^{a_\alpha b_\alpha}) + \frac{1}{2}t_{a_\beta b_\beta}^{i_\beta j_\beta} (g_{i_\beta j_\beta}^{a_\beta b_\beta} - g_{j_\beta i_\beta}^{a_\beta b_\beta}) + t_{a_\alpha b_\beta}^{i_\alpha j_\beta} g_{i_\alpha j_\beta}^{a_\alpha b_\beta} \quad (1.22)$$

1.2 Spin-restricted non-relativistic MP2 energy

The spin-restricted expression for MP2 energy can be obtained from Eq. (1.22) by simply dropping the spin indices as

$$E_{\text{RMP2}} = \frac{1}{2} t_{ab}^{ij} (g_{ij}^{ab} - g_{ji}^{ab}) + \frac{1}{2} t_{ab}^{ij} (g_{ij}^{ab} - g_{ji}^{ab}) + t_{ab}^{ij} g_{ij}^{ab} \quad (1.23)$$

$$= t_{ab}^{ij} (2g_{ij}^{ab} - g_{ji}^{ab}) \quad (1.24)$$

1.3 Kramers restricted relativistic MP2 energy

If P denotes a spinor (Kramers up), then let $\bar{P}$ denote¹ the time reversed version (Kramers down) of P , i.e.,

$$|P\rangle = \mathcal{T}|\bar{P}\rangle; \quad \{|P\rangle\} \cup \{|\bar{P}\rangle\} = \{|p\rangle\} \quad (1.25)$$

where $\mathcal{T}$ is the time reversal operator:

$$\mathcal{T} = \begin{cases} i\sigma_y \hat{K}_0, & \text{2-component} \\ i\Sigma_y \hat{K}_0, & \text{4-component} \end{cases} \quad (1.26)$$

and $\hat{K}_0$ is the complex conjugation operator.

Due to the symmetry of the matrix elements under time reversal, the following relationships arise for electron repulsion integrals for closed shell systems:

$$g_{PQ}^{RS} = g_{\bar{P}\bar{Q}}^{*\bar{R}\bar{S}} = g_{\bar{P}\bar{S}}^{*\bar{R}Q} = g_{RQ}^{*P\bar{S}} \quad (1.27)$$

$$-g_{\bar{Q}\bar{R}}^{*S\bar{P}} = -g_{RQ}^{*PS} = g_{QP}^{\bar{S}\bar{R}} = g_{PQ}^{R\bar{S}} = -g_{\bar{S}\bar{R}}^{*\bar{Q}\bar{P}} = -g_{\bar{R}\bar{S}}^{*\bar{P}\bar{Q}} = g_{\bar{S}\bar{P}}^{*Q\bar{R}} = g_{\bar{P}\bar{S}}^{*\bar{R}Q} \quad (1.28)$$

$$g_{\bar{R}\bar{S}}^{*PQ} = g_{PQ}^{\bar{R}\bar{S}} \quad (1.29)$$

$$g_{\bar{P}\bar{Q}}^{*R\bar{S}} = g_{PQ}^{\bar{R}\bar{S}} \quad (1.30)$$

Let's rewrite the MP2 energy expression, separating it into each Kramers pair permuted blocks,

$$E_{\text{MP2}} = \frac{1}{4} \left[t_{AB}^{IJ} \bar{g}_{IJ}^{AB} \right. \quad (1.31)$$

$$+ \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}\bar{B}} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{A\bar{B}} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}B} \quad (1.32)$$

$$+ \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}\bar{B}} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{A\bar{B}} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}B} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{A\bar{B}} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}\bar{B}} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}B} \quad (1.33)$$

$$+ \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}\bar{B}} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{A\bar{B}} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}B} + \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}\bar{B}} \quad (1.34)$$

$$+ \bar{t}_{AB}^{IJ} \bar{g}_{IJ}^{\bar{A}\bar{B}} \left. \right] \quad (1.35)$$

¹Note: $\bar{}$ over an index denotes time reversal and $\bar{}$ over an operator denotes antisymmetrization